EAGER: TRIAGE - Trustworthy Run-efficient Investigation of Agentic Guided Experiments

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Target Program: Operations Engineering (OE) | **Secondary Program:** Advanced Manufacturing (AM)
Proposal Type: EAGER

Program Relevance: This project develops analytical methods for improving operations in complex, decision-driven engineering environments by embedding statistical experimental design into agentic AI. The methods will be first evaluated through computer experiments across three engineering-operations applications and then transferred to physical manufacturing examples to assess sim-to-real performance.

Key Terms: agentic AI; digital twins; experimental design; loop engineering; quality engineering

Concept Outline

Overview. Loop engineering assigns an agent a real optimization problem and allows it to experiment independently. The agent proposes a change, evaluates it against a metric, retains or discards the change, and repeats the process. This approach aligns naturally with engineering operations, where optimizing a process, monitoring scheme, or production configuration can involve a sequence of costly experiments. However, the practice is based on an untested premise. Agents often appear to explore the design space in a naive manner, resembling random or greedy search rather than principled experimentation.

Overarching hypothesis. The central hypothesis is that classical design of experiments (DOE), when provided as structured guidance to an AI agent, enables the agent to: (a) achieve superior engineering solutions in fewer experimental runs and (b) accumulate transferable experimental knowledge that is distinct from its pretrained priors. To evaluate this hypothesis, we will pursue the following specific aims:

Aim 1. Construct computer simulations with known ground truth to evaluate if DOE-structured guidance enhances within-problem experimentation. We explore three domains for our simulations: machine learning/predictive maintenance, process monitoring, and simulation-based optimization of production tasks. We vary the guidance from a bare goal (improve the metric) to a full DOE protocol.

Aim 2. Evaluate whether the agent transfers acquired knowledge to new problems. We compare two agentic conditions: (a) beginning a new problem with previously acquired knowledge (warm-start) and (b) starting without prior knowledge (cold-start). We compare the reported effects to the true effects of each simulation. We hypothesize that a warm-start will reduce the number of experimentation steps.

Aim 3. Demonstrate transfer to at least one physical manufacturing task. We apply the approach to a manufacturing problem, e.g., 3-D bin packing or robotic assembly. We hypothesize that DOE-structured guidance similarly enhances a physical task. As a proof of concept, we take an initial step toward narrowing the observed gap between simulated and physical environments (sim-to-real gap) in practice.

Intellectual Merit. We frame agentic experimentation as a design-of-experiments problem. Our goal is not to improve the agent, but to structure the experiments it conducts. This framing lets us leverage the semantic reasoning of LLMs. Standard tuning approaches used in machine learning (e.g., Optuna and Bayesian optimization) lack this semantic capability and return only a tuned configuration. However, a DOE-structured AI agent can capture what it learns, reuse that knowledge across problems, and reason about which factors matter. We span multiple models/harnesses to test our approach's generalizability.

Broader Impacts. Our work can result in: (a) democratizing rigorous experimentation by providing non-experts interpretable, transferable insight rather than black-box tuning; (b) improving the efficiency of agentic AI, fewer runs save compute far beyond engineering; (c) translating agentic experimentation from standard computer science problems to engineering domains; and (d) training students at the DOE and AI interface while releasing our simulations and templates as open community infrastructure.